

NAME: _____
ChE 210

Quiz #1

17 September 2025

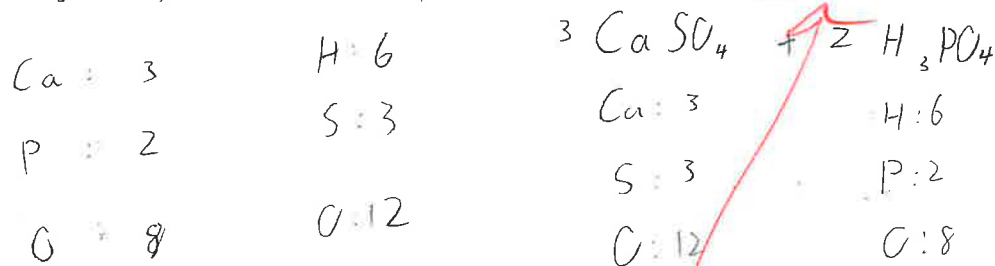
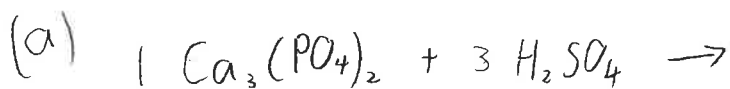
To Receive Full Credit, You Must Show All Your Work.

Be sure to use units correctly in all of your calculations.

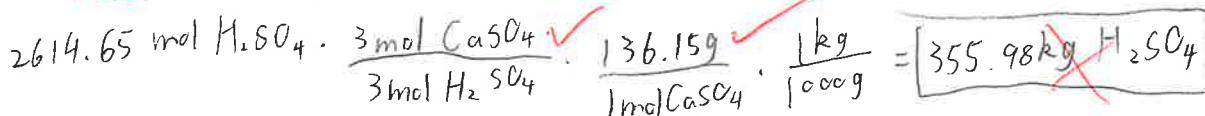
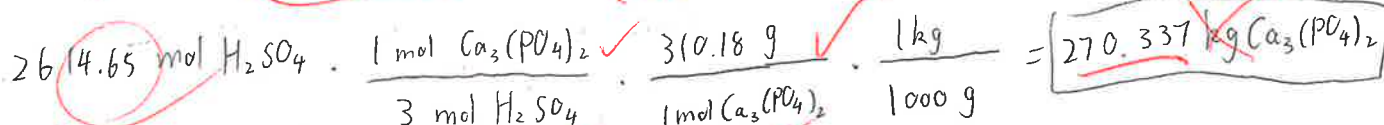
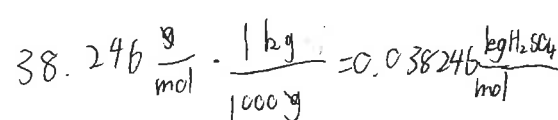
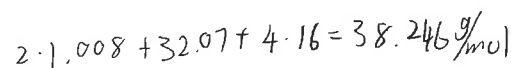
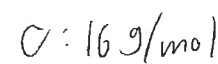
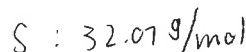
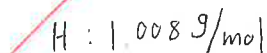
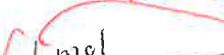
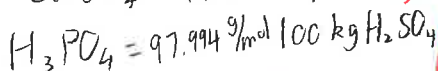
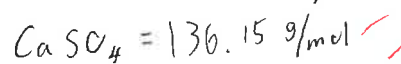
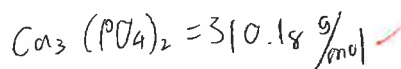
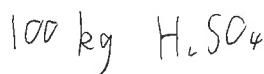
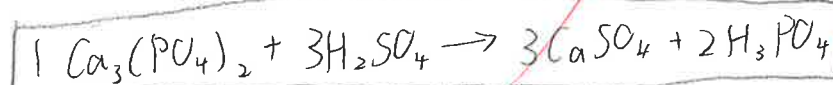
Phosphoric acid (H_3PO_4) can be produced by reacting calcium phosphate ($\text{Ca}_3(\text{PO}_4)_2$) with a stoichiometric quantity of sulfuric acid (H_2SO_4). Calcium sulfate (CaSO_4) is also a byproduct of this reaction.

- (a) Write out the balanced chemical reaction equation. Make all of the stoichiometric coefficients the smallest whole numbers possible. Use any method you want to balance the reaction equation.
- (b) For 100 kg of H_2SO_4 , determine the kg of calcium phosphate consumed and the kg of phosphoric acid and calcium sulfate produced.
- (c) If calcium phosphate costs \$950/ metric-ton and sulfuric acid costs \$113/ metric-ton, and if phosphoric acid sells for \$960/ metric-ton and calcium sulfate sells for \$385/ metric-ton, what is the net profit or loss rate (\$/day) for the process based on selling 250 metric-ton/day of phosphoric acid produced only? Recalculate the profit/loss rate for this process considering that calcium sulfate can also be sold.
- (1 metric-ton = 1000 kg)

Species	Molar Mass
$\text{Ca}_3(\text{PO}_4)_2$	310.18
H_2SO_4	98.08
H_3PO_4	97.994
CaSO_4	136.14



Use $\text{Ca}_3(\text{PO}_4)_2$ as basis, Ca 3, so $3 \cdot (\text{CaSO}_4)$, then balance out the rest.



$$2614.65 \text{ mol } H_2SO_4 \cdot \frac{2 \text{ mol } H_3PO_4}{3 \text{ mol } H_2SO_4} \cdot \frac{97.994 \text{ g}}{1 \text{ mol } H_3PO_4} \cdot \frac{1 \text{ kg}}{1000 \text{ g}} = 170.81 \text{ kg } H_3PO_4$$

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(c)

$\frac{\$950}{1 \text{ ton } Ca_3(PO_4)_2}$	$\frac{\$113}{1 \text{ ton } H_2SO_4}$	$\frac{\$960}{1 \text{ ton } H_3PO_4}$	$\frac{\$385}{1 \text{ ton } CaSO_4}$
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$$\frac{250 \text{ mton}}{\text{day}}$$

$$- 950 \cdot 250 = \$237500 / \text{day } Ca_3(PO_4)_2$$

$$- 113 \cdot 250 = 28250 / \text{day } H_2SO_4$$

$$960 \cdot 250 = 240000 / \text{day } H_3PO_4$$

$$385 \cdot 250 = 96250 / \text{day } CaSO_4$$

$$- 237500 + 28250 = -265750 \text{ ?}$$

$$- 240000 + 96250 = -336250 \text{ ?}$$

$$336250 - 265750 = \$70500 / \text{day} \text{ ?}$$

not right, 250 is only for H_3PO_4 , not enough

time? need to use the balanced equation to find all other values (how much needed/produced), then individual rates to find value. NO!

You already had all the numbers you needed. You are not using a scale factor. That's not a question of time.